Predictable-source Riesz geometry, weighted Schur improvement, and the low-margin boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

first issued 2026-07-30 · this version issued 2026-07-30 · v1.0

1. Purpose and scope

R-126 differentiates the complete R-125 variance-minus-forest endpoint and extracts its six-row spatial Euler force. It also states the causal firewall: that raw force is generally one filtration level too late and is not itself a legal source-space Riesz vector. This note resolves the abstract source geometry exactly and then asks how far it advances the production bound.

There are five proved advances.

1. The adjoint of a temporally faithful source map is computed on the predictable subspace. Conditional expectation occurs separately in every covariance block. The quotient Cameron–Martin Riesz vector and its norm are therefore explicit.
2. A two-block finite fixture shows that replacing this blockwise formula by the unrestricted covariance expression $C_J G$ creates an exact factor-two error.
3. A weighted Schur argument improves both R-126 Hilbert–Schmidt shell constants. At the first collar the mixed coefficient decreases from $0.2981423969\ldots$ to $0.2672612419\ldots$.
4. The two-channel Loewner test is extended by the low/injected channel. At saturation, a noncancelling low coupling has no finite constant payment. A strict operator margin is therefore mandatory unless the complete endpoint supplies an exact weighted cancellation.
5. Normalized Gibbs–Doob information is shown unable to determine an absolute endpoint boundary because it is invariant under a constant gauge. A coherent residual interpolation gives the exact replacement curvature coordinate, but its linear low/injected anchor must remain.

These statements do not identify the complete R-126 force with a bounded production source vector. In particular, R-125 does not yet prove that the probability-root/spatial-shell decomposition is the block decomposition of one common terminal source Hessian. No production forward/legal-reverse/ balanced/low estimate, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson estimate, cutoff or floor removal, interacting measure, Sector-A closure, or tier promotion is claimed. The host claim remains T4.

The setting is a fixed finite cutoff, fixed positive floor, and a fixed R-093/R-104 temporally faithful chart. All Hilbert spaces below are real; the complex formulation is obtained by inserting adjoints and real parts.

2. Exact predictable-source adjoint and quotient Riesz vector

Let the chart have time blocks $b = 1, \ldots, B$. Put

$$\mathcal{S}_\pi^{\mathrm{pred}} = \bigoplus_{b=1}^{B} L^2(\Omega, \mathcal{F}_{t_{b-1}}; \mathcal{S}_b), \qquad L_\pi h = \sum_{b=1}^{B} S_b h_b, \tag{2.1}$$

where $S_b = (\Delta C_b)^{1/2}$ and S_b is measurable at the corresponding strict-past boundary. The physical Cameron–Martin space is the completion of $\operatorname{Ran} L_\pi$ in the quotient norm

$$\|u\|_{q,L_\pi} = \inf\{\|h\|_{\mathcal{S}_\pi} : L_\pi h = u\}. \tag{2.2}$$

Theorem 2.1 (legal source adjoint)

For every square-integrable physical dual field G,

$$\boxed{(L^*_{\pi,\mathrm{pred}} G)_b = \mathbb{E}[S_b^* G \mid \mathcal{F}_{t_{b-1}}].} \tag{2.3}$$

Proof. For predictable h_b, conditional expectation and strict-past measurability give

$$\begin{aligned}\mathbb{E}\langle G, L_\pi h\rangle &= \sum_b \mathbb{E}\langle S_b^* G, h_b\rangle \\ &= \sum_b \mathbb{E}\langle \mathbb{E}[S_b^* G \mid \mathcal{F}_{t_{b-1}}], h_b\rangle.\end{aligned} \tag{2.4}$$

This is precisely the defining adjoint identity on the closed predictable source subspace. □

Write

$$g_b(G) = \mathbb{E}[S_b^* G \mid \mathcal{F}_{t_{b-1}}]. \tag{2.5}$$

Since $g = L^*_{\pi,\mathrm{pred}} G$ belongs to $\overline{\operatorname{Ran} L^*_{\pi,\mathrm{pred}}} = (\operatorname{Ker} L_\pi)^\perp$, the physical quotient Riesz vector is

$$\boxed{\mathcal{R}^{\mathrm{CM}}_\pi(G) = L_\pi g = \sum_b S_b \mathbb{E}[S_b^* G \mid \mathcal{F}_{t_{b-1}}],} \tag{2.6}$$

and its quotient norm is exact:

$$\|\mathcal{R}^{\mathrm{CM}}_\pi(G)\|_{q,L_\pi} = \|g(G)\|_{\mathcal{S}_\pi}. \tag{2.7}$$

Let $C_J = \sum_b S_b S_b^*$. Conditional Jensen gives the sharp general upper bound

$$\begin{aligned}\|g(G)\|^2_{\mathcal{S}_\pi} &= \sum_b \mathbb{E}\|\mathbb{E}[S_b^* G \mid \mathcal{F}_{t_{b-1}}]\|^2 \\ &\le \sum_b \mathbb{E}\|S_b^* G\|^2 = \mathbb{E}\langle G, C_J G\rangle.\end{aligned} \tag{2.8}$$

The constant in (2.8) is one. The inequality need not be equality because predictability removes a different future component in every block.

Combining (2.8) with the R-120 dual covariance coercivity gives, within that theorem's declared symbol class,

$$\|g(G)\|_{\mathcal{S}_\pi} \le \sqrt{M_R^2/c_{\mathrm{sym}}}\,\|G\|_{H^{-2}}, \qquad \sqrt{M_R^2/c_{\mathrm{sym}}} = 3.0377807308\ldots. \tag{2.9}$$

This is an abstract dual bound. It becomes a production estimate only after the complete R-126 force is proved to lie in the R-120 dual symbol class with uniform constants.

Counterfixture 2.2 (the unrestricted covariance collapse)

Take two scalar blocks with $S_1 = S_2 = 1$. Let $\mathcal{F}_{t_0}$ be trivial, $\mathcal{F}_{t_1} = \sigma(\xi)$, and $\xi = \pm 1$ with equal probability. For $G = \xi$,

$$g_1 = \mathbb{E}\xi = 0, \qquad g_2 = \mathbb{E}[\xi \mid \sigma(\xi)] = \xi. \tag{2.10}$$

Consequently

$$\mathcal{R}^{\mathrm{CM}}_\pi(G) = \xi, \qquad C_J G = 2\xi. \tag{2.11}$$

The legal quotient norm squared is one, while $\mathbb{E}\langle G, C_J G\rangle = 2$. Thus the upper bound (2.8) is correct, but the identity $\mathcal{R}^{\mathrm{CM}}_\pi(G) = C_J G$ is false. Moving the predictable projection outside the covariance-block sum loses its time label and creates an exact factor-two error in this smallest fixture.

3. Straight homotopies and the source completion cost

Let $\mathfrak{F}$ be a differentiable expected endpoint functional with physical gradient $G(Z)$, and let

$$Z_t = X_J + tL_\pi h, \qquad \overline{G}_h = \int_0^1 G(Z_t)\,\mathrm{d}t. \tag{3.1}$$

Then the fundamental theorem of calculus and Theorem 2.1 give

$$\mathfrak{F}(Z_1) - \mathfrak{F}(Z_0) = \sum_b \mathbb{E}\langle h_b, g_b(h)\rangle, \qquad g_b(h) = \mathbb{E}[S_b^*\overline{G}_h \mid \mathcal{F}_{t_{b-1}}]. \tag{3.2}$$

Moreover,

$$\|g(h)\|^2_{\mathcal{S}_\pi} \le \int_0^1 \mathbb{E}\langle G(Z_t), C_J G(Z_t)\rangle\,\mathrm{d}t. \tag{3.3}$$

Same-root visits may represent the line integral in (3.2), but the terminal displacement must be compressed before taking the quotient norm. A sum of visit squares is not representation invariant.

At the production value $q = 10/9$, the source cross has the exact completion

$$\boxed{\langle h, g\rangle + \frac{9}{20}\|h\|^2 = \frac{9}{20}\left\|h + \frac{10}{9}g\right\|^2 - \frac{5}{9}\|g\|^2.} \tag{3.4}$$

Hence the adverse force-square coefficient is exactly 5/9. No argument that uses the source energy only can spend a smaller universal coefficient after the same cross term has been isolated.

4. The production interface and the common-Hessian requirement

For probability root r, same-root visit v, and output shell m, R-126 gives the complete six-row Euler force

$$\begin{aligned} G_r(W) = 2\sum_{a=1}^{6}\sum_i \{&\mathbb{E}_{f,r}[(Dc_a)^T\Delta\Gamma_{r,i}c_a] + \bar{c}_{a,r}\,\partial_i\Phi_{a,i,r} \\ &+ \Phi_{a,i,r}\mathbb{E}_{f,r}[(Dc_a - Dc_a^T)\partial_i W]\}. \end{aligned} \tag{4.1}$$

Here $\mathbb{E}_{f,r} = \mathbb{E}[\,\cdot \mid \mathcal{F}_{r-1} \vee \sigma(\xi_r)]$. Rows, output clusters, heat, the once-owned covariance, the R-063 forest, and the Cartan contribution must be assembled before the output projection Π_m. Only after this future conditioning is complete may the source projection $\mathbb{E}[\,\cdot \mid \mathcal{F}_{t_{b-1}}]$ be applied.

If $A_{r,v;b}$ is the exact derivative of the endpoint with respect to source block b, the candidate legal source coordinate is

$$g_b = \sum_{r,v,m}\int_0^1 \mathbb{E}[A^*_{r,v;b}\Pi_m G_r(W_{r,v,t}) \mid \mathcal{F}_{t_{b-1}}]\,\mathrm{d}t. \tag{4.2}$$

Formula (4.2) is the correct incidence pattern, not yet a proved production identity. Same-root subdivision must first compress to the common terminal displacement.

The obstruction can also be stated at Hessian level. For $\Phi = \mathbb{E}_f[c(W)^T V]$, define

$$\begin{aligned} \dot\Phi[H] &= \mathbb{E}_f[(Dc\,H)^T V + c^T\partial H], \\ \ddot\Phi[H,K] &= \mathbb{E}_f[(D^2c[H,K])^T V + (Dc\,H)^T\partial K + (Dc\,K)^T\partial H]. \end{aligned} \tag{4.3}$$

For the R-126 total symbol,

$$\begin{aligned} D^2\mathcal{T}_r[H,K] = &\sum_i \mathbb{E}_f\int D^2B[H,K] : \Delta\Gamma_{r,i}\,\mathrm{d}x \\ &- 2\sum_{a,i}\mathrm{Re}\int\{\dot\Phi_{a,i}[H]\overline{\dot\Phi_{a,i}[K]} + \Phi_{a,i}\overline{\ddot\Phi_{a,i}[H,K]}\}\,\mathrm{d}x. \end{aligned} \tag{4.4}$$

The pulled source Hessian

$$\Pi_{\mathrm{ad}} L_\pi^* D^2 \mathcal{T}_r L_\pi \Pi_{\mathrm{ad}} \tag{4.5}$$

is quotient-basic and selfadjoint once (4.5) is genuinely one common endpoint/source-space object. The Cartan coefficient survives in the selfadjoint first-order form

$$A_i \partial_i + \frac{1}{2} \partial_i A_i, \qquad A_i^* = -A_i, \tag{4.6}$$

including the registered 2680/729 recombined coefficient. Formal selfadjointness does not make a rectangular probability-root/spatial-shell reverse band the adjoint of the forward band. That conclusion requires the missing common-terminal source–source factorization.

5. A weighted Schur theorem for triangular shell kernels

Theorem 5.1 (weighted triangular Schur bound)

Let $0 < d \le 1/2$ and $q^2 < 1 - d$. On nonnegative indices define

$$B_{pj} = d^j q^{p-j} \mathbf{1}_{\{j \le p\}}. \tag{5.1}$$

Then

$$\boxed{\|B\|_{\ell^2 \to \ell^2} \le \left(1 - \frac{q^2}{1-d}\right)^{-1/2}.} \tag{5.2}$$

Proof. Use Schur weights $w_j = t^j$ with $t = q/(1-d)$. The normalized row sum is

$$R_p = \frac{1}{w_p} \sum_{j \le p} B_{pj} w_j = \sum_{j=0}^{p} d^j (1-d)^{p-j}. \tag{5.3}$$

It obeys $R_0 = 1$ and

$$R_{p+1} = (1-d)^{p+1} + d R_p \le (1-d) + d = 1 \tag{5.4}$$

by induction. The normalized column sum is

$$\frac{1}{w_j} \sum_{p \ge j} B_{pj} w_p = \frac{d^j}{1 - qt} \le \frac{1}{1 - q^2/(1-d)}. \tag{5.5}$$

The weighted Schur test gives the square root of the product of (5.4) and (5.5), which is (5.2). □

Corollary 5.2 (mixed production prototype)

Suppose the as-yet-unproved production blocks satisfy

$$\|A_{\mathrm{mix},mr}\| \le C_{\mathrm{mix}} 2^{r-2m}, \qquad r \ge j_0, \quad m \ge r + C. \tag{5.6}$$

Put $r = j_0 + j$ and $m = j_0 + C + p$; then $j \le p$ and (5.6) is the kernel (5.1) with $d = 1/2$, $q = 1/4$, multiplied by $C_{\mathrm{mix}} 2^{-j_0 - 2C}$. Therefore

$$\boxed{\|A_{\mathrm{mix}}\| \le C_{\mathrm{mix}} 2^{-j_0 - 2C} \sqrt{8/7}.} \tag{5.7}$$

At $C = 1$, the coefficient is

$$\frac{1}{4}\sqrt{8/7} = \frac{1}{\sqrt{14}} = 0.267261241912424\ldots, \tag{5.8}$$

strictly below the R-126 Hilbert–Schmidt coefficient $2/\sqrt{45} = 0.298142396999972\ldots$.

The retained R-126 operator budget is

$$K_{\text{bud}} = 0.920932333907528\ldots. \tag{5.9}$$

Thus the collar-one mixed-only ceiling, when no other same-block channel spends the budget, improves from

$$C_{\text{mix}} 2^{-j_0} < \frac{\sqrt{45}}{2} K_{\text{bud}} = 3.08890095194215\ldots \tag{5.10}$$

to

$$\boxed{C_{\text{mix}} 2^{-j_0} < \sqrt{14} K_{\text{bud}} = 3.44581326988407\ldots.} \tag{5.11}$$

This is an $11.5546702\ldots\,\%$ improvement in the admissible mixed coefficient. If a far block occupies the same source and target channels, the sufficient triangle condition is instead

$$\frac{\sqrt{8/7}}{4} C_{\text{mix}} 2^{-j_0} + \frac{\sqrt{224/223}}{16} C_{\text{far}} 2^{-3j_0} \le K_{\text{bud}}, \tag{5.12}$$

or one must verify the complete signed block-Loewner matrix directly.

Corollary 5.3 (far production prototype)

If instead

$$\|A_{\text{far},mr}\| \le C_{\text{far}} 2^{r-4m}, \tag{5.13}$$

the same indexing gives $d = 1/8$ and $q = 1/16$. Hence

$$\boxed{\|A_{\text{far}}\| \le C_{\text{far}} 2^{-3j_0 - 4C} \sqrt{224/223}.} \tag{5.14}$$

At $C = 1$ its coefficient is $0.0626399777789193\ldots$, below the old Hilbert–Schmidt value $8/\sqrt{16065} = 0.0631174757774868\ldots$.

These are conditional operator theorems: they improve the aggregation once the block estimates (5.6) or (5.13) are proved. They do not prove those production estimates. Mixed and far blocks occupying the same source and target channels add by the triangle inequality; no quadrature is used without an orthogonality theorem.

6. The augmented low/injected Loewner criterion

Let X, Y, H be Hilbert spaces. Assume that $R = R^*$ on X, $S = S^*$ on Y, and $L = L^*$ on H are bounded, and that $A : Y \to X$, $B : H \to X$, and $C : H \to Y$ are bounded. Let the adverse real quadratic form be

$$\begin{aligned} q(x, y, \ell) = {} & \langle x, Rx \rangle + \operatorname{Re}\langle x, Ay \rangle + \langle y, Sy \rangle \\ & + 2\operatorname{Re}\langle x, B\ell \rangle + 2\operatorname{Re}\langle y, C\ell \rangle + \langle \ell, L\ell \rangle. \end{aligned} \tag{6.1}$$

The source and sextic payments $2\eta\|x\|^2 + 2\zeta\|y\|^2$ dominate (6.1) for every (x, y, ℓ) if and only if

$$\boxed{\begin{pmatrix} 2\eta I - R & -A/2 & -B \\ -A^*/2 & 2\zeta I - S & -C \\ -B^* & -C^* & -L \end{pmatrix} \succeq 0.} \tag{6.2}$$

Put $D = -L$ and

$$M_2 = \begin{pmatrix} 2\eta I - R & -A/2 \\ -A^*/2 & 2\zeta I - S \end{pmatrix}, \qquad K = \begin{pmatrix} B \\ C \end{pmatrix}. \tag{6.3}$$

The bounded-operator Schur complement says that (6.2) is equivalent to

$$D \succeq 0, \qquad \operatorname{Ran} K^* \subseteq \operatorname{Ran} D^{1/2}, \qquad M_2 - X_D^* X_D \succeq 0, \tag{6.4}$$

where X_D is the Douglas reduced solution of $D^{1/2}X_D = K^*$. If D has closed range, then $X_D = (D^\dagger)^{1/2}K^*$ and the last condition is equivalently $M_2 - KD^\dagger K^* \succeq 0$. Thus (6.4) does not silently assume a bounded Moore–Penrose inverse when the range is not closed. Thus an unbudgeted low block must have $L \preceq 0$. If $L = 0$, positivity of every two-by-two principal compression forces $B = C = 0$. An unsigned low term cannot merely be declared "retained"; it requires a negative reserve, an explicit constant budget, or exact cancellation.

For the affine scalar low remainder, write

$$M_0 = \begin{pmatrix} 2\eta I - R & -A/2 \\ -A^*/2 & 2\zeta I - S \end{pmatrix}, \qquad d = \begin{pmatrix} b \\ c \end{pmatrix}. \tag{6.5}$$

The residual

$$\langle z, M_0 z\rangle - 2\operatorname{Re}\langle d, z\rangle + C_0 - \ell \tag{6.6}$$

is nonnegative for every $z = (x, y)$ if and only if

$$M_0 \succeq 0, \quad d \in \operatorname{Ran} M_0^{1/2}, \quad C_0 - \ell \geq \|x_d\|^2, \tag{6.7}$$

where x_d is the Douglas reduced solution of $M_0^{1/2}x_d = d$. If M_0 has closed range, then $\|x_d\|^2 = \langle d, M_0^\dagger d\rangle$. When $R = S = 0$, all variables are scalar, $\eta, \zeta > 0$, and $a^2 < 16\eta\zeta$,

$$C_{0,\min} - \ell = \frac{2\zeta b^2 + abc + 2\eta c^2}{4\eta\zeta - a^2/4}. \tag{6.8}$$

At the two-channel saturation $a = 4\sqrt{\eta\zeta}$, the null vector is $(\sqrt{\zeta}, \sqrt{\eta})$. The constant in (6.7) is finite precisely when

$$\boxed{b\sqrt{\zeta} + c\sqrt{\eta} = 0.} \tag{6.9}$$

Therefore saturating the old two-by-two Loewner budget leaves no room for a generic low/injected affine coupling.

If

$$|a| \leq (1-\theta)4\sqrt{\eta\zeta}, \qquad 0 < \theta \leq 1, \tag{6.10}$$

then

$$\begin{aligned} \lambda_{\min}(M_0) &= \eta + \zeta - \sqrt{(\eta-\zeta)^2 + a^2/4} \\ &\geq \frac{2\eta\zeta(2\theta - \theta^2)}{\eta + \zeta}. \end{aligned} \tag{6.11}$$

The last line follows from the determinant lower bound and $\lambda_{\max}(M_0) \leq \operatorname{Tr} M_0$. This gives a quantitative place to pay a uniform dual norm for the low/injected remainder, if such a production norm is proved.

7. The common-terminal middle is a coordinate, not a reserve

Under the R-079 common-terminal and matching-trace hypotheses, write

$$f_j = d_j(J_j - J_0), \qquad i_j = d_j(J_* - J_j), \qquad m_j = f_j + i_j = d_j(J_* - J_0). \tag{7.1}$$

Then the sum of the forward and injected blocks is

$$F_j + I_j = \mathbb{E}\left[\langle d_j J_0, m_j\rangle + \frac{1}{2}\|m_j\|^2\right] - \frac{1}{2}\mathbb{E}\Delta\tau_j^{*0}. \tag{7.2}$$

It depends only on the two endpoints and not on the chosen middle J_j.

Choosing the artificial Doob middle $\widetilde{J}_j = J_0 + P_j(J_* - J_0)$ can make a displayed reverse coordinate zero. Restoring the low block returns the full endpoint difference $\mathbb{E}[W(A_*) - W(A_0)]$. The Nelson burden has only moved coordinates.

Predictability also gives no sign. Let G be a revealed scalar random variable and put

$$J_0 = 0, \quad J_1 = \tanh G, \quad J_* = (1+c)\tanh G. \tag{7.3}$$

The second insertion $c \tanh G$ is strict-past after the first reveal, yet

$$I_1 = \mathbb{E}[\tanh^2 G]\left(c + \frac{c^2}{2}\right). \tag{7.4}$$

It is positive for $c > 0$ and negative for $-2 < c < 0$. This is an abstract legal filtration fixture, not a full A1 production counterexample.

8. The normalized Gibbs–Doob absolute-anchor no-go

For an endpoint energy L, normalized Gibbs laws depend on

$$\frac{e^{-qL}}{\mathbb{E}e^{-qL}}. \tag{8.1}$$

Replacing L by $L+C$ leaves (8.1), all normalized conditional laws, Doob increments, conditional variances, and relative entropies unchanged. The absolute free energy

$$\Phi(L) = -\frac{1}{q}\log \mathbb{E}e^{-qL} \tag{8.2}$$

instead satisfies

$$\Phi(L+C) = \Phi(L) + C. \tag{8.3}$$

Therefore no theorem using only normalized Gibbs–Doob or entropy data can bound the absolute low/free-energy endpoint. An absolute anchor from the low/injected ledger is logically indispensable. Thermodynamic integration and the KL chain rule reproduce the same endpoint gap and do not supply an independent reserve.

9. A coherent residual interpolation

The gauge obstruction suggests interpolating the complete residual before normalization. Fix the base measure and stabilizer throughout the interpolation, and take R and $\delta\tau$ independent of s. Let

$$Y = b + A\xi + R, \qquad Y_s = b + A\xi + sR, \qquad \tau_s = \mathrm{Tr}(AA^*) + s\delta\tau, \tag{9.1}$$

and define

$$P_s = \frac{\|Y_s\|^2 - \tau_s}{2}, \qquad \Psi(s) = \log \int e^{-qP_s}\,\mathrm{d}\lambda. \tag{9.2}$$

With μ_s the normalized tilted law,

$$\dot{P}_s = \mathrm{Re}\langle Y_s, R\rangle - \frac{\delta\tau}{2}, \qquad \ddot{P}_s = \|R\|^2, \tag{9.3}$$

and direct differentiation gives

$$\boxed{\Psi''(s) = q^2\,\mathrm{Var}_{\mu_s}(\dot{P}_s) - q\mathbb{E}_{\mu_s}\|R\|^2.} \tag{9.4}$$

The exact integral Taylor formula is

$$\Psi(1) - \Psi(0) = \Psi'(0) + \int_0^1 (1-s)\Psi''(s)\,\mathrm{d}s, \qquad \Psi'(0) = -q\mathbb{E}_{\mu_0}\dot{P}_0. \tag{9.5}$$

The linear anchor in (9.5) is generically nonzero. It must be summed with the complete low/injected endpoint before any variance argument. The live production target is therefore a root-summed integrated variance-minus-curvature estimate for the fully assembled cluster/forest/ Cartan packet, with the absolute anchor retained once.

10. Evidence map and successor lemma

The proof-route decisions are as follows.

Route	Verdict	Reusable reason
Unrestricted $C_J G$ collapse	failed	Predictable projection is blockwise; the two-block fixture has exact factor two.
Weighted Schur aggregation	advanced	Time-triangular weights improve both mixed and far constants without orthogonality.
Saturated two-channel Loewner	failed generically	A noncancelling low coupling is outside the saturated matrix range.
Eliminate the reverse middle	inconclusive	The common-terminal endpoint is independent of the chosen middle.
Normalized Gibbs/entropy anchor	failed	Constant gauges preserve normalized data and shift absolute free energy.
Coherent residual curvature	live	The exact variance-minus-curvature identity retains the required first-order anchor.

The smallest noncyclic successor is now precise:

$$\boxed{P_{\text{pred}} L_\pi^* D(\text{complete R-093 endpoint action}) = g^{\text{adm}}_{\text{low/inj}} + \sum_{r,m} g^{\text{adm}}_{r,m}.} \tag{10.1}$$

It must be proved with the $\Delta C/\Delta\Gamma$ normalization, heat, exhaustive clusters, R-063 forest, rational multipliers, 2680/729 Cartan, trace and covariance each exactly once, and invariance under representation-preserving refinement. The resulting augmented Loewner matrix must have either the strict margin (6.11) or the exact cancellation (6.9). Only then may the R-093 directed chart union be invoked toward $\mathrm{OVERLAP}_{\mathrm{src}}$ and Nelson.

11. Devil's-advocate review

1. **Objection:** $C_J G$ already has the right covariance units, so inserting every conditional projection is cosmetic. **UPHELD against the shortcut.** Counterfixture 2.2 produces $C_J G = 2\mathcal{R}_\pi^{\mathrm{CM}}(G)$. Equation (2.6), not $C_J G$, is the legal Riesz identity.

2. **Objection:** the improved Schur constants may hide a changed shell normalization. **DISMISSED.** The factors 2^{-j_0-2C} and 2^{-3j_0-4C} are taken outside before applying the dimensionless kernels $(d, q) = (1/2, 1/4)$ and $(1/8, 1/16)$. At $C = 1$ they are compared with the R-126 constants in the same normalization.

3. **Objection:** far and mixed improvements may be combined in quadrature. **UPHELD unless orthogonality is separately proved.** Shared source/target channels require the triangle inequality.

4. **Objection:** the old two-channel saturation already spends the exact available budget, so the low term can be kept outside the matrix. **UPHELD against that separation.** Equations (6.7)–(6.9) show that a generic low coupling has infinite affine cost at saturation.

5. **Objection:** normalized Doob squares or KL production supply the missing absolute constant. **DISMISSED.** The constant-gauge transformation leaves all those data fixed while shifting (8.2).

6. **Objection:** formal selfadjointness identifies the legal reverse band automatically. **UPHELD as a missing hypothesis.** It is valid only after the rectangular blocks are shown to arise from the single source Hessian (4.5).

7. **Objection:** the executable hardcodes upstream derived reserves. **UPHELD against the draft and repaired.** Both routes read the exact operator budget from the manifest-pinned R-126 primary result; all new radicals are derived from it, with decimals used only as labelled regression intervals.

8. **Objection:** finite matrix checks prove the infinite Schur theorem only numerically. **DISMISSED.** Equations (5.3)–(5.5) are the analytic infinite-dimensional proof; finite matrices are only executable regression fixtures.

9. **Objection:** the strict-margin premise may be one-sided in the mixed scalar a. **UPHELD against the draft and repaired.** Equation (6.10) uses $|a|$, and both executable routes test invariance under $a \mapsto -a$.

10. **Objection:** Moore–Penrose formulas are automatically bounded for every positive diagonal block. **UPHELD against the draft and repaired.** Both (6.4) and (6.7) use bounded Douglas reduced solutions; the Moore–Penrose forms are asserted only for closed range.

12. Reproducibility

Primary exact audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_predictable_source_riesz_weighted_schur_
  low_margin_boundary.py
```

Non-importing independent audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_predictable_source_riesz_weighted_schur_
  low_margin_boundary_independent.py
```

Integrated authority, PDF, and public-surface audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_predictable_source_riesz_weighted_schur_
  low_margin_boundary_verify.py
```

13. Result footer

```
Result ID: A13-CLASSII-PREDICTABLE-SOURCE-RIESZ-WEIGHTED-
  SCHUR-LOW-MARGIN-BOUNDARY (R-127).
Precise statement: Theorems 2.1 and 5.1; source completion in Section 3;
  common-Hessian boundary in Section 4; augmented Loewner criterion in
  Section 6; Gibbs gauge and coherent curvature boundaries in Sections 8-9.
```

Scope: fixed finite cutoff and positive floor; fixed R-093/R-104
temporally faithful chart; square-integrable predictable source blocks;
fixed base measure/stabilizer and s-independent residual data in Section 9;
production shell applications conditional on their stated block prototypes.
Dependencies: A1; R-079, R-093, R-104, R-120, R-121, R-125, R-126.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_predictable_source_riesz_weighted_
schur_low_margin_boundary_verify.py
Expected output: R-127 integrated PASS; primary 52/52, independent 60/60,
manifest, PDF, exploration, negative, and public-surface gates pass.
Falsification gate: any failed predictable-adjoint or quotient identity,
factor-two fixture, q=10/9 completion, Schur domain/constant, exponent
relabel, augmented Loewner/range/strict-margin test, gauge or curvature
identity, authority/hash, PDF, exploration, or public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no complete production source-Hessian
identification, projected-force norm, unified forward/legal-reverse/
balanced/low closure, OVERLAP_src, Nelson, removals, interacting measure,
Sector-A closure, or T5-T7 follows.
Next required action: prove the common complete-endpoint/source-Hessian
identity with every owner present once, then prove its augmented
forward/legal-reverse/balanced/low Loewner bound before chart union.